Xiaoyan (Elena) Bai

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RESEARCH INTERESTS

My research investigates how models work internally and how those mechanisms shape their observable behavior, with the goal of enabling reliable scientific discovery.

EDUCATION

University of Chicago **September 2024 – Present**
Ph.D. in Computer Science *Chicago, Illinois, United States*

- Advisor: Prof. Chenhao Tan

University of Michigan, Ann Arbor **September 2022 – April 2024**
B.S.E. in Computer Science, Minor in Art and Design *Ann Arbor, Michigan, United States*

- Relevant coursework: Machine Learning, Natural Language Processing, Computer Vision, and Computer Game Design and Development

Shanghai Jiao Tong University **September 2020 – August 2024**
B.S. in Electronic and Computer Engineering *Shanghai, China*

PUBLICATIONS

ARA: Agent-Native Research Artifacts *AI4Research and AI4Science Workshops at ICML 2026* (**Spotlight**)
Jiachen Liu, Jiaxin Pei, Jintao Huang, Chenglei Si, Ao Qu, Xiangru Tang, Runyu Lu, Lichang Chen, **Xiaoyan Bai**, et al.

Mechanistic Interpretability of Chemical Language Models: Molecular Geometry and Representations in MoLMFormer and SMI-TED *Mechanistic Interpretability Workshop at ICML 2026*
Wanqing Hao, Zetong Xu, Zhiyuan Han, **Xiaoyan Bai**, Chenhao Tan

The Story is Not the Science: Execution-Grounded Evaluation of Mechanistic Interpretability Research *AI4Science Workshop at ICML 2026* (**Best Poster Award**)
Xiaoyan Bai, Alexander Baumgartner*, Haojia Sun*, Ari Holtzman, Chenhao Tan

Know Thyself? On the Incapability and Implications of AI Self-Recognition *preprint, MSLD 2026*
Xiaoyan Bai, Aryan Shrivastava, Ari Holtzman, Chenhao Tan

Why Can't Transformers Learn Multiplication? Reverse-Engineering Reveals Long-Range Dependency Pitfalls *preprint*
Xiaoyan Bai*, Itamar Pres*, Yuntian Deng, Chenhao Tan, Stuart Shieber, Fernanda Viégas, Martin Wattenberg, Andrew Lee

"You Are a Brilliant Mathematician" Does Not Make LLMs Act Like One *Agents4Science 2025*
AI Agents, **Xiaoyan Bai**, Ari Holtzman, Chenhao Tan

Concept Incongruence: An Exploration of Time and Death in Role Playing. *NeurIPS 2025, MMLS 2025* **(Best Paper Honorable Mention)**
Xiaoyan Bai, Ike Peng*, Aditya Singh*, Chenhao Tan

Learn To be Efficient: Build Structured Sparsity in Large Language Models. *NeurIPS 2024* (**Spotlight [Top 2% of submissions]**)
Haizhong Zheng, **Xiaoyan Bai**, Xueshen Liu, Z. Morley Mao, Beidi Chen, Fan Lai, Atul Prakash

A Mechanistic Understanding of Alignment Algorithms: A Case Study on DPO and Toxicity. *ICML 2024* (**Oral [Top 1.5% of submissions]**)
Andrew Lee, **Xiaoyan Bai**, Itamar Pres, Martin Wattenberg, Jonathan K. Kummerfeld, Rada Mihalcea

INDUSTRIAL EXPERIENCE

Research Intern **June 2026 - September 2026**
Copilot Tuning, Microsoft *California, United States*

- Built a transferable, agentic data-preparation pipeline that trains models to generalize to unseen downstream tasks.
- Designed an agentic pipeline to mitigate shortcut learning and enable scalable transfer of reasoning capabilities.

AI Training Assistant Intern **December 2021 – January 2022**
Emogent *Shanghai, China*

- Preprocessed and analyzed training data in Python for "Irene," an interactive AI museum guide.
- Expanded the system's knowledge base and improved its conversational capabilities.

RESEARCH EXPERIENCE

MATS Scholar **2026**
Advisor: Alan Cooney *MATS Program*

- Developed a lie-confession training method to improve the safety of AI systems.
- Conducted experiments to characterize incoherent deceptive behavior in model organisms.

Undergraduate Research Assistant **2022 – 2024**
Advisor: Prof. Rada Mihalcea and Dr. Andrew Lee *University of Michigan*

- Examined alignment algorithms by tracing the circuits reinforced by direct preference optimization, showing how they can lead to brittle behaviors.
- Developed a tool to detect toxicity in the internal representations of language models.
- Preprocessed poetry data and fine-tuned a vision-language model to generate poems from images.

Undergraduate Research Assistant **2023 – 2024**
Advisor: Prof. Atul Prakash and Dr. Haizhong Zheng *University of Michigan*

- Experimented with sparsity in models to improve inference efficiency in GeLU-based LLMs.
- Developed grouping methods to convert dense models into mixtures of experts, improving robustness and efficiency.
- Analyzed activation patterns and attention mechanisms to guide token-pruning decisions.

TEACHING EXPERIENCE

Teaching Assistant, *Math Foundations for ML*, University of Chicago **October 2024 – December 2024**

Grader, *Foundations of Computer Science*, University of Michigan **September 2023 – December 2023**

Teaching Assistant, *Game Design*, Massachusetts Institute of Technology Lincoln Laboratory **June 2023 – August 2023**

Teaching Assistant, *Intro to Engineering*, Shanghai Jiao Tong University **September 2021 – August 2022**

PROFESSIONAL SERVICE

Organizer, Interpretability for Discovery Workshop at NeurIPS 2026 **2026**

Conference Reviewer **2024 – 2026**

- Transactions on Machine Learning Research (TMLR)
- COLM 2026
- ICML 2026 (**Gold Reviewer**)
- ICLR 2026; DeLTa and SCOPE Workshops 2025
- NeurIPS 2025; Mechanistic Interpretability Workshop 2025
- ACL Rolling Review (ARR) 2024

VOLUNTEERING

Code to PhD: Volunteer **2024 – Present**

HONORS & AWARDS

ICML Gold Reviewer **2026**

Shanghai Jiao Tong University Excellent Graduate **2024**

University of Michigan Dean's Honor List **2023 – 2024**

China Collegiate Algorithm Design and Programming Challenge Contest – Bronze Medal **March 2021**

Shanghai Jiao Tong University Undergraduate Scholarship **September 2020 – August 2021**